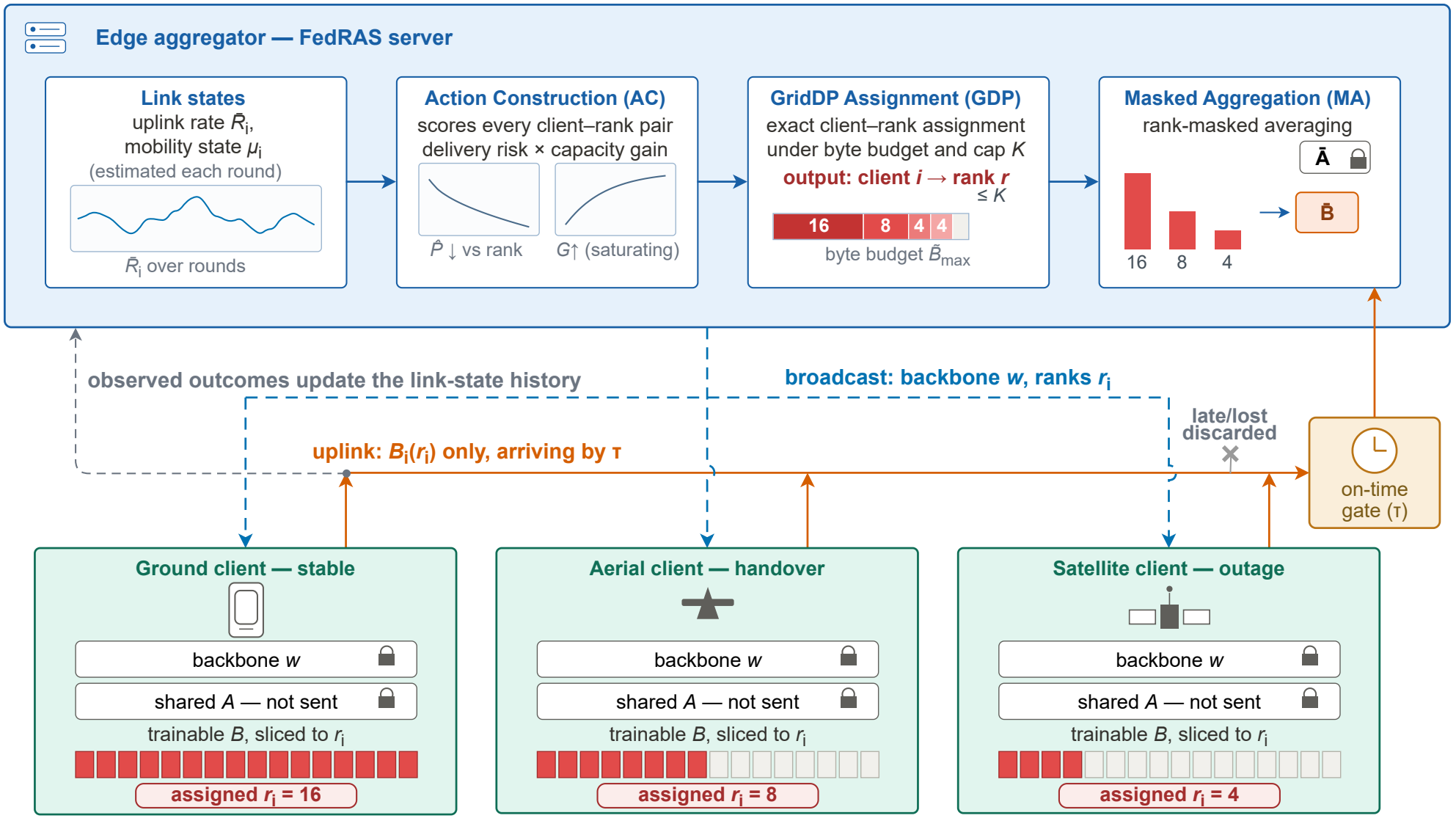

Edge aggregator — FedRAS server
Link states
uplink rate $\bar{R}_i$,
mobility state μ_i
(estimated each round)
$\bar{R}_i$ over rounds
Action Construction (AC)
scores every client–rank pair
delivery risk × capacity gain
$\hat{P}\downarrow$ vs rank
$G\uparrow$ (saturating)
GridDP Assignment (GDP)
exact client–rank assignment
under byte budget and cap K
output: client $i \rightarrow$ rank $r \leq K$
16 8 4 4
byte budget $\tilde{B}_{max}$
Masked Aggregation (MA)
rank-masked averaging
$\bar{A}$
$\bar{B}$
16 8 4
observed outcomes update the link-state history
broadcast: backbone w, ranks r_i
uplink: $B_i(r_i)$ only, arriving by τ
late/lost
discarded
on-time
gate (τ)
Ground client — stable
backbone w
shared A — not sent
trainable B, sliced to r_i
assigned $r_i = 16$
Aerial client — handover
backbone w
shared A — not sent
trainable B, sliced to r_i
assigned $r_i = 8$
Satellite client — outage
backbone w
shared A — not sent
trainable B, sliced to r_i
assigned $r_i = 4$